Taylor Polynomials and Remainder Bounds

AP Calculus BC — Convergence and Taylor Series

Building a Taylor polynomial from derivative values, building one by substituting into a known Maclaurin series, bounding the remainder, and using a polynomial to approximate a value or a limit. Decide which of the four you are being asked for before you write anything:

- **From derivatives.** $T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$. The derivative value is the numerator; the factorial is always there.
- **From a known series.** Substitute into a memorised Maclaurin series (e^u, $\sin u$, $\cos u$, $\ln(1+u)$, $\frac{1}{1-u}$) rather than differentiating four times.
- **Lagrange bound.** $|R_n(x)| \le \frac{M\,|x-c|^{n+1}}{(n+1)!}$, where M is any number with $|f^{(n+1)}(z)| \le M$ for every z between c and x. The exponent and the factorial both use $n+1$, never n.
- **Alternating series bound.** If the series alternates with terms decreasing in size to 0, the error is smaller than the *first omitted term* — and its sign tells you whether the estimate is high or low.

Leave answers exact (fractions, not decimals) unless a problem says otherwise.

1. A function f is four times differentiable and its derivatives at $x = 2$ are given below.

$f(2)$	$f'(2)$	$f''(2)$	$f'''(2)$
3	-1	4	12

Write the third-degree Taylor polynomial T_3 for f about $x = 2$, then use it to approximate $f(2.5)$.

Answer: ________________

2. Write the fourth-degree Maclaurin polynomial for $f(x) = e^x$, then use it to approximate e (that is, evaluate the polynomial at $x = 1$). Give your answer as an exact fraction.

Answer: ________________

3. The Maclaurin series for $\sin x$ is $x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$

Use its fifth-degree Maclaurin polynomial to approximate $\sin\left(\frac{1}{2}\right)$. Give an exact fraction.

Answer: ________________

4. Starting from the Maclaurin series for e^u, write the sixth-degree Maclaurin polynomial for $f(x) = e^{-x^2}$ (do not differentiate — substitute $u = -x^2$), then use it to approximate $f\left(\frac{1}{2}\right)$. Give an exact fraction.

Answer: ________________

5. The third-degree Maclaurin polynomial for e^x is used to approximate $e^{0.5}$.

(a) Taking $M = 2$ as a bound for $\left|f^{(4)}(z)\right|$ on $[0, 0.5]$, use the Lagrange error bound to bound $|R_3(0.5)|$. Give an exact fraction.

(a) ________________

(b) Justify why $M = 2$ is a legitimate choice on this interval.

6. Replace e^x by its Maclaurin series to evaluate

$$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$$

Show the series step — do not use L'Hôpital's rule.

Answer: ________________

7. The fourth-degree Maclaurin polynomial for $\cos x$ is used to approximate $\cos\left(\frac{2}{5}\right)$. The Maclaurin series for $\cos x$ alternates, so the error is smaller than the first omitted term. Write that bound as an exact fraction.

Answer: ________________

8. The Maclaurin series for $\ln(1+x)$ is $x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots$
Use its first four nonzero terms to approximate $\ln(1.2)$. Give an exact fraction.

Answer: ____________________

9. A function g satisfies $\left|g^{(5)}(z)\right| \leq 40$ for every z in $[1, 1.5]$. The fourth-degree Taylor polynomial for g about $x = 1$ is used to approximate $g(1.5)$.
Bound the error $|R_4(1.5)|$. Give an exact fraction.

Answer: ____________________

10. Synthesis challenge. Let $f(x) = \ln(1 + x)$.

(a) Write the third-degree Maclaurin polynomial T_3 for f, and use it to approximate $\ln(1.1)$. Give an exact fraction.

(a) ______________

(b) The series for $\ln(1 + x)$ alternates with terms decreasing in size. Use the alternating series error bound to bound the error in your part (a) estimate. Give an exact fraction.

(b) ______________

(c) Is the part (a) estimate an overestimate or an underestimate of $\ln(1.1)$? Justify your answer from the sign of the first omitted term, not by computing $\ln(1.1)$ on a calculator.